

ABDALLAH SHEE

Full-Stack Software Engineer — Node.js · Express · Next.js · TypeScript · PostgreSQL · Python (Data & Automation)

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Portfolio: <https://portfolio-git-main-abdallahshees-projects.vercel.app/>

PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in Node.js, Express.js, and Next.js with TypeScript used end-to-end across client, server, and database layers, backed by PostgreSQL for scalable, type-safe data modeling and Python for data manipulation, file processing, and workflow automation. Experienced building production-ready SaaS, fintech, and regulatory-compliance systems, with strong command of API design, authentication and authorization, middleware architecture, and performance-tuned backend systems that scale.

TECHNICAL SKILLS

Core Specialization

- **TypeScript** — advanced, end-to-end across client, server, and database layers; type-safe contracts throughout the stack
- **Node.js & Express.js** — RESTful and real-time API design, middleware pipelines, modular service architecture
- **Next.js** — App Router, SSR/SSG/ISR, Server Actions, Edge Middleware
- **PostgreSQL** — schema design, type-safe queries with Drizzle ORM, migrations, performance tuning, optimized search/filtering for high-traffic platforms
- **Python** — data manipulation and cleaning, file processing, automation scripts, preprocessing pipelines for downstream analytics and ML workflows

Supporting Frameworks & Data Layer

- Frontend: React, Redux, TanStack Query, TanStack Start
- Additional Data Stores: MongoDB with Mongoose ODM (schema modeling, aggregation, indexing); Redis and AWS ElastiCache for caching and reducing database load

Auth, Cloud & DevOps

- Authentication & Authorization: session/token-based auth, OAuth flows, role-based access control (RBAC), Supabase Auth, secure middleware and request/response validation
- Cloud & Serverless: Vercel, AWS Lambda, edge middleware, horizontally scalable stateless service design, CDN/edge-caching strategies
- Tooling & Collaboration: Git/GitHub (branching, PR workflows, GitHub Actions/CI-CD), npm/pnpm, Agile cross-functional collaboration
- AI-Assisted Development: agentic coding tools including Claude Code for code generation, refactoring, debugging, and test scaffolding

PROFESSIONAL EXPERIENCE

Full-Stack Developer — Quantum Plus Ltd

Jun 2023 – Sep 2025

Mombasa, Kenya

- Built and maintained full-stack, production-grade applications using Next.js, Node.js, Express, TypeScript, and PostgreSQL, supporting multi-user SaaS and e-commerce platforms.
- Designed and developed secure, type-safe REST APIs with Express.js middleware layers and scalable backend architecture to support business growth.
- Implemented role-based access control (RBAC) to power secure, multi-tenant SaaS and e-commerce functionality across distinct user roles and permission levels.
- Drove testing, debugging, and PostgreSQL query/performance optimization to ensure system reliability, data integrity, and compliance across customer-facing features.

- Streamlined frontend state management and data-fetching workflows using Redux, Zustand, and TanStack Query, improving UI responsiveness and reducing boilerplate across the codebase.

Data Analyst — *CloudFactory Kenya*

Feb 2022 – Mar 2023

Nairobi, Kenya

- Used Python to process, clean, and structure large-scale datasets, building repeatable automation scripts that powered reliable downstream software and analytical workflows.
- Designed and built Python-based data pipelines and preprocessing scripts to support programmatic data handling and machine learning workflows.
- Developed and tested predictive models and algorithms in Python following scalable, maintainable coding practices, strengthening foundations in data-driven, production-quality engineering.

Junior Web Developer — *Swahilipot Hub Foundation*

2020 – 2022

Mombasa, Kenya

- Built responsive, interactive UI components using React and TypeScript, translating UI/UX designs into functional, cross-browser-compatible code.
- Collaborated closely with designers and backend developers to integrate APIs and deliver cohesive end-to-end features.
- Managed database integration with PostgreSQL and MongoDB to support reliable CRUD operations across core application features, laying the groundwork for full-stack development expertise.

KEY PROJECTS

- **eTIMS Integration System** — Developed invoicing systems in Node.js/Express and TypeScript integrating with Kenya's eTIMS API, implementing backend validation and role-based access control to ensure regulatory compliance and production reliability through rigorous testing and debugging.
- **Booking Platforms & SaaS Applications** — Built multi-role SaaS booking platforms in Next.js and PostgreSQL with complex scheduling logic, advanced search and filtering, and role-based dashboards; leveraged Redis caching to speed up search/filter queries and reduce database load.
- **E-Commerce Platforms** — Developed product catalog, shopping cart, checkout, and admin/user interfaces with integrated inventory and order management features using Next.js, Express, and PostgreSQL.

EDUCATION

Bachelor of Business Information Technology

Dedan Kimathi University of Science and Technology, Nyeri | 2014 – 2018

Hands-on experience in programming, web development, database management, and the software development lifecycle, with foundational knowledge in business information systems and IT infrastructure.

CERTIFICATIONS

- Nanodegree in Cloud Computing (AWS) — Udacity, 2023
- Certificate in Financial Technology (FinTech) — Corporate Finance Institute (CFI), May 2023
- Certificate in Business Intelligence — Corporate Finance Institute (CFI), Nov 2022
- Certificate in Software Development — Power Learn Project Academy, Nairobi, Jan 2022 – Apr 2022